Sabari Vishnu Jayanthan Jaikrishnan

Staff AI / ML Engineer | Staff Data Scientist

Ontario, Canada | Canadian Permanent Resident | Authorized to work in Canada without sponsorship

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Professional Summary

Staff-level AI/ML engineer and data scientist with 7 years of experience building production-grade RAG, agentic AI, and computer vision systems. Technical lead for multi-site AI/data platform delivery across global operations, translating complex operational data into decision-ready insights and taking systems from prototype to measurable production impact.

Career Highlights

- Served as technical lead for a multi-site AI/data platform, driving production output from 30K to 32K parts/day, and delivered ML risk models mitigating ~$12M in annual losses.
- Architected and shipped production RAG systems — LangChain, pgvector, hybrid vector+BM25 retrieval, human-feedback tuning — cutting enterprise knowledge-discovery time from 40+ hours to minutes.
- Built agentic AI systems using MCP tool servers with controlled, retrieval-grounded tool execution, reducing hallucination and unsafe-action risk while recommending optimized machine parameters in production.

Technical Skills

- **Product Analytics / Data Science:** Deep Learning, Computer Vision, Predictive ML, Gradient Boosting, Random Forest, Time Series Forecasting, Classification, Regression, Clustering, A/B Testing, Anomaly Detection, Risk Detection
- **Data / Analytics:** Apache Kafka, PostgreSQL, pgvector, MongoDB, MySQL, Dataiku, Power BI, Tableau, TIBCO Spotfire, Matplotlib
- **Programming:** Python, SQL, JavaScript, React, Node.js, Flask, FastAPI, REST APIs, Microservices
- **ML Frameworks:** TensorFlow, PyTorch, pandas, NumPy, scikit-learn
- **GenAI:** Generative AI, Agentic AI, RAG pipelines, LangChain, prompt engineering, hallucination mitigation, OpenAI text-embedding-3, hybrid search (vector + BM25), semantic chunking, vector search, Claude API, OpenAI API, MCP
- **Cloud / MLOps:** AWS, AWS Lambda, Docker, Kubernetes, Jenkins CI/CD, OpenVINO, edge inference (Intel Movidius NCS), model deployment, production monitoring, Git

Leadership & Delivery

Owned end-to-end architecture for cross-functional AI/data initiatives — from problem framing through production adoption — partnering with operations, engineering, and leadership stakeholders, and influencing leadership decisions on AI/data infrastructure investment based on platform outcomes.

Professional Experience

Staff Data Scientist | Western Digital (WD)

January 2021 – Present

- Architected a cloud-native RAG platform (AWS Lambda, Claude API, LangChain, pgvector) that indexed thousands of documents across SharePoint, wikis, and engineering reports using structure-aware chunking and hybrid vector+BM25 retrieval. Reached **100+ users**, with SME-feedback-driven accuracy tracking via Power BI, and cut enterprise knowledge-discovery time from **40+ hours to minutes**.
- Designed the architecture for a real-time AI-driven decision-optimization platform, built in Python with MCP tool servers and pgvector for controlled tool execution, then containerized it with Docker and Kubernetes across multiple production lines to **recommend optimized machine parameters within seconds** while reducing hallucination and unsafe-action risk.
- Served as technical lead for a multi-site AI/data platform spanning 2 global manufacturing sites, standardizing reusable ML workflows, deployment patterns, and monitoring practices — improving delivery velocity and system reliability while driving a **6.7% increase in production throughput** (30K→32K units/day).
- Trained a real-time computer vision model (TensorFlow SSD) on NVIDIA GPUs and deployed inference at the edge via OpenVINO 2024 on an Intel Movidius Neural Compute Stick, with HLS streaming and a Python/React/Node.js stack, to monitor real-time resource utilization — **improving operational efficiency by 16%** and contributing to a World Economic Forum Global Lighthouse recognition.
- Productionized a gradient boosting risk-detection pipeline, orchestrated through Jenkins CI/CD, to identify hidden defect patterns tied to **~$12M in annual losses**, shifting decision-making from reactive analysis to proactive risk mitigation.
- Mentored 3-5 junior and intermediate engineers on ML practices including model evaluation, testing, and versioning through code and design reviews, driving consistent production ML deployment across teams.

Systems Engineer | Infosys Pvt. Ltd. (Client: U.S.-Based Automotive Enterprise)

June 2016 – June 2018

- Led **migration of a legacy billing system** to Oracle SOA BPEL on Oracle Fusion Middleware while embedded on a U.S. enterprise client account, integrating SQL, XML, and REST APIs for production-grade data access.
- Built **reusable data pipelines** and warehouse integration workflows, improving data consistency across transactional systems and downstream reporting.
- **Worked directly with client business and IT stakeholders**, translating requirements into BPEL process designs and supporting delivery scope and timelines.

Education

Master of Science in Information and Data Science

King Mongkut's University of Technology North Bangkok, Thailand – 2020 | CGPA: 4.0 / 4.0

Bachelor of Technology in Electronics and Communication Engineering

SRM University, India – 2016 | CGPA: 8.5 / 10